The closure-response operator $C_\varphi = L_M + \varphi^{-2} I$: a geometry-fixed kernel on the 600-cell with two domain-disjoint empirical witnesses

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Headline. We define a programme-level closure-response primitive $C_\varphi = L_M + \varphi^{-2} I$ on a closure substrate M with corresponding Laplacian L_M (graph, simplicial, or continuum) and golden ratio $\varphi = (1+\sqrt{5})/2$. We use the 600-cell instance V_{600} as the discrete substrate shared by the two empirical witnesses (the choice of this polytope is post-hoc motivated by those landings, §9; numerically reproduced: $|V| = 120$, $|E| = 720$, uniform degree 12, H_3 shell decomposition $\{1, 12, 20, 12, 30, 12, 20, 12, 1\}$, computed Laplacian spectrum matching the closed-form $\mathbb{Z}[\varphi]$ values), establish the operator-norm bound $\|C_\varphi^{-1}\| = \varphi^2$ from the smallest eigenvalue φ^{-2}, and verify the discrete-to-continuum agreement at per-vertex Pearson correlation 0.976 between the discrete Green response and the continuum kernel $G(x) = (\varphi/2)\, e^{-|x|/\varphi}$ (this is a multiplicity-weighted shell-radius correlation, since ψ is shell-constant to machine precision under H_3-fixed source; not 119 independent radial samples; §5; `repro/verify_kernel.py`). The same fixed C_φ on the same fixed graph is then the shared response primitive used underneath two *domain-disjoint* empirical works: a passive-regime structural fit of the $b \to s\mu^+\mu^-$ angular anomaly across five public flavour-physics datasets [Smart, 2026c], and an active-regime substrate witness against eighteen preregistered substrate/neuroscience correspondences plus six drug/sleep EEG signatures [Smart, 2026b].

Scope. This paper presents an empirical *operator witness*: a geometry-fixed response operator with operator shape held fixed across two disjoint-domain empirical landings (flavour physics and cortical neuroscience) with no kernel-shape retuning between regimes; the substrate V_{600} is post-hoc disclosed as the choice motivated by these landings, not selected by an independent criterion. It is *not* a derivation of the φ^{-2} shift from first principles, *not* a uniqueness claim for V_{600} among regular 4-polytopes, *not* a selection theorem on the companion adaptive-closure-transport 4-tuple [Smart, 2026a], and *not* a model-preference claim against alternative kernels on either empirical landing (the b-anomaly AIC comparison and the aria-chess substrate-witness scope are documented in their own preprints and inherited verbatim here). The positive claim is therefore deliberately narrow but non-trivial: a single fixed shifted-Laplacian response operator on the 600-cell gives a reproduced geometry-only discrete-to-continuum agreement and serves, without kernel-shape retuning, as the common response primitive beneath two domain-disjoint empirical analyses. The significance is not that either empirical domain is solved, but that the same geometry-fixed response object reappears without being reshaped for either domain.

Abstract

We define a closure-response primitive $C_\varphi = L_M + \varphi^{-2} I$ on a closure substrate M with corresponding Laplacian L_M and $\varphi = (1+\sqrt{5})/2$, give the 600-cell graph V_{600} as

the discrete instance shared by two empirical witnesses, and document its appearance as the same fixed operator (no shape retuning) in two domain-disjoint empirical works: (i) a five-dataset structural fit of the $b \to s\mu^+\mu^-$ angular anomaly with sign-uniform amplitude direction [Smart, 2026c]; (ii) an eighteen-prediction preregistered substrate witness against substrate/neuroscience signatures plus six drug/sleep EEG signatures [Smart, 2026b]. We include the numerical reproduction script (`repro/verify_kernel.py`) that constructs V_{600} from canonical generators, verifies the graph facts ($|V| = 120$, $|E| = 720$, uniform degree 12, 9-shell decomposition, Laplacian spectrum numerically matching closed-form $\mathbb{Z}[\varphi]$ values, operator-norm identity $\|C_\varphi^{-1}\| = \varphi^2 \approx 2.618$ on the connected finite graph V_{600} where $\lambda_{\min}(L_M) = 0$), and tests the discrete-to-continuum agreement at multiplicity-weighted shell-radius Pearson correlation 0.976 for the unweighted variant (the discrete response is shell-constant to machine precision), above the two φ-cocycle weighted variants tested (0.888 geometric, 0.884 arithmetic). Within the three tested variants the unweighted Laplacian ranks highest on the geometry-only criterion, reproducing the qualitative ranking established separately by b-anomaly's data χ^2 comparison (the b-anomaly preprint flags that its data was looked at first and the geometry ranking verified afterward; the agreement is criterion-independent but historically non-blind, a caveat we inherit verbatim).

(i) Operator definition and properties. $C_\varphi = L_M + \varphi^{-2} I$ is positive definite for self-adjoint non-negative L_M on a connected finite graph; smallest eigenvalue $\varphi^{-2} \approx 0.382$, operator norm $\|C_\varphi^{-1}\| = \varphi^2 \approx 2.618$. The continuum projection in one coordinate x is the closed-form Green's function $G(x) = (\varphi/2)\, e^{-|x|/\varphi}$, with decay scale φ.

(ii) The 600-cell instance. V_{600} has 120 canonical unit vectors on S^3 generated by three orbits (8 axis, 16 half-integer, 96 φ-mixed). H_4 transitivity forces uniform degree 12 on the short-edge graph; the Laplacian has nine eigenvalue classes in $\mathbb{Z}[\varphi]$ with multiplicities summing to 120 (Table 1, §3).

(iii) Discrete-to-continuum agreement. Multiplicity-weighted per-vertex Pearson correlation (shell-constant response, see §5) between the discrete response $\psi = C_\varphi^{-1} f$ for a localised source and the continuum prediction $G(\|v - v_{\mathrm{src}}\|)$ at each non-source vertex's chord distance: 0.976 (unweighted Laplacian), 0.888 (φ-geometric weights), 0.884 (φ-arithmetic weights). Same source vertex, same fixed shift, no parameter fitting; the unweighted-headline correlation is invariant under choice of source to within machine precision over all 120 vertices (multi-source sweep, §9; the sweep is run on the unweighted operator only, the two weighted variants were not swept).

(iv) Two domain-disjoint empirical witnesses. (a) Passive regime, b-anomaly [Smart, 2026c]: same C_φ on the same V_{600} provides a sign-uniform $\Delta C_9^{\mathrm{eff}} < 0$ description of the $b \to s\mu^+\mu^-$ angular anomaly across five public datasets (LHCb 2015, LHCb 2021, CMS 2025, LHCb 2025, LHCb 2015 $B_s \to \phi$), with one fitted dimensionless amplitude per dataset and the kernel shape held fixed. (b) Active regime, aria-chess [Smart, 2026b]: same C_φ on the same V_{600}, augmented by a recurrent self-model layer with one condition-dependent self-injection coupling $\eta \in \{0, 0.05, 0.20\}$ and one substrate-pinned nonlinearity $\mathrm{bounded\_topk}(\cdot, k{=}12)$ at the graph's average degree, passes 17/18 preregistered correspondences (frozen 2026-04-18) at standard methodology and 18/18 after a documented $N = 20$ deep-dive on the residual high-variance $C \times P$ interaction (P4) — alongside two further documented methodology refinements in the aria-chess set ($N = 5$ for P3 within its preregistered band; LOO/state-reset for the P13 chess substrate-lift, originally preregistered with 5-fold CV; no threshold modifications) and one open prereg-exact rerun (P10 chess null mapping was validated with 15 permutations rather than the preregistered 20; threshold $\geq 50\%$ unchanged; 15–20-range robust at 65.4%; the 20-permutation rerun is left open in the aria-chess limitations) — plus 6/6 literature-thresholded drug/sleep EEG signatures on a single deterministic seed.

What we do not claim. We do not derive the φ^{-2} floor; it is a design-level stability clamp making C_φ strictly positive definite. We do not claim V_{600} is the unique substrate consistent with either empirical landing. We do not claim the operator is the unique kernel shape consistent with the b-anomaly data (b-anomaly's free-width Gaussian alternative and Mode-B refit drift caveats are inherited verbatim) nor that the aria-chess substrate witness establishes a selection theorem on the adaptive-closure-transport 4-tuple. The structural scope of this paper is: *one geometry-fixed operator on one fixed graph appears as the shared response primitive underneath two domain-disjoint empirical works in two distinct regimes, with no shape retuning between them.*

1 Introduction

A response operator on a fixed graph, with no shape parameters tuned to any dataset, that simultaneously gives (i) a structural fit consistent with the q^2 shape of the $b \to s\mu^+\mu^-$ angular anomaly across five public flavour-physics datasets in their passive linear-response regime, and (ii) eighteen preregistered substrate/neuroscience correspondences plus six drug/sleep EEG signatures in the active dynamical regime of a recurrent self-model layer above the same graph, deserves a separate preprint that names the operator, gives its construction in full, and threads the relationship between the two empirical landings without inheriting either's load-bearing claims. That is what this paper does.

The operator is

$$C_\varphi \;=\; L_M + \varphi^{-2} I, \qquad \varphi \;=\; (1+\sqrt{5})/2, \tag{1}$$

where M is a closure substrate (graph, simplicial complex, or projected coordinate) and L_M is its Laplacian. The shift $\varphi^{-2} \approx 0.382$ regularises the inverse: for self-adjoint non-negative L_M on a connected finite graph, C_φ is strictly positive definite, the smallest eigenvalue is φ^{-2}, and the operator-norm bound is

$$\|C_\varphi^{-1}\| \;=\; 1/\varphi^{-2} \;=\; \varphi^2 \;\approx\; 2.618. \tag{2}$$

The continuum projection in one coordinate x has a closed-form Green's function $G(x) = (\varphi/2)\, e^{-|x|/\varphi}$ with decay scale φ (§2).

The discrete substrate used by the two empirical witnesses is the 600-cell graph V_{600}: 120 unit vectors on S^3, generated by three standard coordinate families (8 axis vertices, 16 half-integer vertices, 96 φ-mixed vertices), connected by short edges $\langle v, w\rangle = \varphi/2$. The choice of this polytope is post-hoc motivated by the empirical landings (§9); the construction itself is theorem-level rigorous. The graph has $|E| = 720$ edges, uniform degree by H_4 transitivity (with the value 12 from the short-edge construction), a 9-shell H_3 partition $\{1, 12, 20, 12, 30, 12, 20, 12, 1\}$, and antipodal symmetry $s(-v) = 8 - s(v)$. The Laplacian spectrum has nine eigenvalue classes in $\mathbb{Z}[\varphi]$ with multiplicities summing to 120. All of these facts are reproduced numerically by `repro/verify_kernel.py` from the canonical generators alone — no external data input.

What this paper claims

We claim a single *operator witness*: that one geometry-fixed operator, on one fixed graph, with no shape-parameter retuning between regimes, appears as the shared response primitive underneath two empirical works covering qualitatively distinct physical settings.

1. **Operator definition is fixed by the construction.** Once V_{600} is selected and the stability shift φ^{-2} is chosen, C_φ is fully determined. No shape parameter, no fitted threshold, no learned weight enters the operator. The Laplacian spectrum, the operator-norm bound, and the discrete-to-continuum agreement are computed (not fitted) from the construction and reproduced in `repro/verify_kernel.py`.

2. **Discrete-to-continuum agreement is empirical, not postulated.** For a localised source at any vertex, the discrete response $\psi = C_\varphi^{-1} f$ is shell-constant to machine precision, and the multiplicity-weighted per-vertex Pearson correlation between $\psi(v)$ and $G(\|v - v_{\mathrm{src}}\|)$ is $\rho = 0.976$ on the unweighted graph Laplacian. (This is a multiplicity-weighted shell-radius correlation, not 119 independent radial samples; the within-shell variance is at machine precision; see §5.2.) It is numerical agreement between two independently-defined objects (a 120-dimensional discrete inverse and a continuum exponential kernel), not a definitional identity.

3. **Variant comparison among the three tested variants.** Two φ-cocycle weighted Laplacian variants (φ-geometric, φ-arithmetic edge weights via shell-grade exponents $\omega_+(v) =$

$\varphi^{\kappa(v)}$) score lower per-vertex correlation: 0.888 and 0.884 respectively. Within the three tested variants, the unweighted Laplacian ranks highest on the geometry-only criterion. This reproduces, on a different test, the qualitative ranking established independently by the b-anomaly paper's data-χ^2 comparison (§6).

4. **Two domain-disjoint empirical landings, same operator.** (a) The b-anomaly preprint [Smart, 2026c] uses the same fixed C_φ on the same V_{600} to describe the q^2 shape of the $b \to s\mu^+\mu^-$ anomaly across five public datasets, with one fitted dimensionless amplitude per dataset and the kernel held fixed; sign uniformity holds in 5/5 datasets ($A > 0$, $\Delta C_9^{\text{eff}} < 0$). (b) The aria-chess preprint [Smart, 2026b] uses the same fixed C_φ on the same V_{600}, augmented by a recurrent self-model layer above the substrate, to pass eighteen preregistered substrate/neuroscience correspondences (frozen 2026-04-18) plus six drug/sleep EEG signatures.

What this paper does *not* claim

- *Not a derivation of the φ^{-2} floor.* The shift φ^{-2} is a design-level stability clamp that bounds $\|C_\varphi^{-1}\|$ at φ^2. It is not derived from a closure functional or a symmetry argument; we report its role as a regularisation-of-mass scale.
- *Not a uniqueness claim for V_{600}.* Other regular 4-polytopes (the 24-cell, the 120-cell), other highly symmetric graphs, and continuum substrates are all candidate M for $C_\varphi = L_M + \varphi^{-2}I$. The 600-cell choice is post-hoc motivated by the empirical landings; a formal ablation against alternative substrates is an open build.
- *Not a kernel-uniqueness claim on either empirical landing.* The b-anomaly's free-width Gaussian alternative shows that a free-width Gaussian charm-loop proxy fits the same five datasets comparably in χ^2 at the cost of one extra shape parameter; the b-anomaly AIC comparison against FREE_C9 (a constant Wilson-coefficient shift) is a statistical tie on current data ($w_{\text{VFD}} = 0.348$ vs $w_{\text{FREE\_C9}} = 0.652$). Both caveats are inherited verbatim from the b-anomaly preprint.
- *Not a selection theorem on the ACT 4-tuple.* The companion adaptive-closure-transport preprint [Smart, 2026a] proposes a selection layer $(M, L_M, W, R_{\text{hom}})$ in which C_φ fills the response slot. Selection — Lyapunov $V(W)$ on the reduced flow, edge-space decomposition under $2I$-equivariance (where $2I$ is the binary icosahedral group, the SU(2) preimage of the icosahedral symmetry of V_{600}), full reduced-flow convergence — is left open in that paper and is not delivered here.
- *Not a circuit-level model on the active-regime side.* The aria-chess preprint operates at the architectural-algorithmic level above C_φ. We import its empirical results verbatim and do not relitigate them here; their substrate-witness scope applies.

Mapping from numerics to admissible claims

To keep this paper inside the operator-witness scope, we use the same claim-boundary discipline as the aria-chess preprint [Smart, 2026b]: numerical results $\mathcal{R}_{\text{numeric}}$ are mapped to admissible claims $\mathcal{C}_{\text{admissible}}$ by the rule

$$q\colon \mathcal{R}_{\text{numeric}} \longrightarrow \mathcal{C}_{\text{admissible}}, \qquad \mathcal{C}_{\text{admissible}} = \{\text{‘computed’, ‘consistent with’, ‘direction confirmed’}\}.$$

A numerically computed quantity (the Laplacian spectrum, the operator-norm bound, the per-vertex correlation 0.976) licenses a ‘computed’-type claim. Downstream empirical agreement using the fixed operator (sign uniformity in 5/5 b-anomaly datasets, the aria-chess 17/18 standard plus 18/18 after the documented $N = 20$ P4 deep-dive) licenses a ‘consistent with’-type claim. We never write ‘derives the kernel’, ‘proves uniqueness’, or ‘establishes selection’.

What is claimed / what is not.
Claimed: a geometry-fixed response operator C_φ on the 600-cell graph, with computed spectrum, operator-norm bound, and discrete-to-continuum correlation; the same fixed operator appearing in two domain-disjoint empirical works (b-anomaly five-dataset structural fit, AIC tie + Mode-B drift caveats inherited; aria-chess 17/18 preregistered correspondences under the standard validation protocol and 18/18 after the documented P4 $N=20$ deep-dive (thresholds unchanged) with the documented P3/P13 methodology refinements and the open prereg-exact P10 20-permutation rerun inherited verbatim, plus 6/6 literature-thresholded companion EEG signatures) without shape-parameter retuning between regimes.
Not claimed: derivation of φ^{-2}; uniqueness of V_{600}; uniqueness of the kernel shape on either empirical landing; a selection theorem on the ACT 4-tuple; that either empirical landing settles the underlying physics (flavour anomaly or consciousness) by the operator alone.

Layout

§2 gives the operator definition, the positivity properties of C_φ, the operator-norm bound, and the continuum projection. §3 constructs V_{600} from canonical generators, gives the graph facts, and the 9-shell decomposition. §4 reports the Laplacian spectrum in $\mathbb{Z}[\varphi]$ with multiplicities, computed numerically. §5 runs the discrete-to-continuum agreement test across three Laplacian variants. §6 and §7 thread the two domain-disjoint empirical witnesses (b-anomaly and aria-chess) at the operator level. §8 positions C_φ relative to the published adaptive-closure-transport companion as a member of the same polynomial-in-L family, and identifies the open selection layer. §9 enumerates limitations in a five-move guard matrix. §10 concludes.

2 The closure-response operator

2.1 Definition

Let M be a closure substrate: a connected finite undirected graph $M = (V, E)$, a finite simplicial complex with chosen Laplacian, or a projected continuum coordinate. Let L_M be the corresponding Laplacian (graph Laplacian $L = D - A$, simplicial Laplacian, or continuum operator $-\Delta$ with chosen boundary conditions). Let $\varphi = (1+\sqrt{5})/2$ be the golden ratio, with $\varphi^{-1} = \varphi - 1$ and $\varphi^{-2} = 2 - \varphi \approx 0.381966$.

The *closure-response operator* is

$$C_\varphi \;=\; L_M + \varphi^{-2} I. \tag{3}$$

For a non-negative localised source f on M, the *closure response field* is

$$\psi \;=\; C_\varphi^{-1} f \;=\; (L_M + \varphi^{-2} I)^{-1} f. \tag{4}$$

2.2 Hypotheses on (M, L_M)

The properties developed in §2.3–§2.4 require:

- **(H1) Self-adjointness.** L_M is self-adjoint on the L^2 inner product on M (with counting measure on a finite graph, with Lebesgue measure with appropriate boundary conditions in the continuum case).
- **(H2) Non-negativity.** $L_M \geq 0$ as a quadratic form: $\langle f, L_M f\rangle \geq 0$ for all f.
- **(H3) Known spectral bottom.** On a finite graph, M is connected (so the kernel of L_M is exactly the constant vector and $\lambda_{\min}(L_M) = 0$). In the continuum case, the spectral bottom $\inf \sigma(L_M)$ is explicitly known on the chosen domain (it equals 0 for the full line; equals the first Dirichlet eigenvalue > 0 on a bounded Dirichlet interval). The full-line case has $\inf \sigma(L_M) = 0$ as spectral bottom but not as an eigenvalue, which is sufficient for the operator-norm identity in §2.4; H3 does not require a finite-dimensional zero eigenspace.

Three concrete settings illustrate the hypothesis class:

- the standard combinatorial Laplacian on a connected finite undirected graph (the 600-cell case, $\lambda_{\min}(L_M) = 0$);
- the continuum $L = -d^2/dx^2$ on the full line with decay-at-infinity (spectral bottom $\inf \sigma(L_M) = 0$, not attained as an eigenvalue; used for the closed-form Green's function in §2.5);
- the continuum $L = -d^2/dx^2$ on a bounded interval with Dirichlet boundary conditions ($\lambda_{\min}(L_M) > 0$).

Substrates outside this class — projected coordinates with non-standard boundaries, weighted Laplacians whose weight function is unbounded, or operators with negative spectrum — require their own analysis, which we do not give here.

2.3 Positive definiteness

Under (H1)–(H3) on a finite connected graph, L_M has a smallest eigenvalue $\lambda_{\min}(L_M) = 0$ with one-dimensional eigenspace (the constant vector). For $C_\varphi = L_M + \varphi^{-2} I$,

$$\lambda_{\min}(C_\varphi) \;=\; \lambda_{\min}(L_M) + \varphi^{-2} \;=\; \varphi^{-2} \;>\; 0,$$

so C_φ is strictly positive definite and C_φ^{-1} is well-defined and bounded.

2.4 Operator-norm bound

The operator norm of C_φ^{-1} is the reciprocal of the spectral bottom of C_φ:

$$\|C_\varphi^{-1}\| \;=\; 1/\inf \sigma(C_\varphi) \;=\; 1/(\inf \sigma(L_M) + \varphi^{-2}). \tag{5}$$

On finite graphs the spectral bottom is the smallest eigenvalue $\lambda_{\min}(L_M)$ (and is attained); on the full-line continuum case (§2.5) it is $\inf \sigma(L_M) = 0$ as spectral bottom but *not* as an eigenvalue. On any substrate where $\inf \sigma(L_M) = 0$ (e.g. the connected finite graph V_{600}, or the full-line continuum case), this reduces to the identity

$$\|C_\varphi^{-1}\| \;=\; \varphi^2 \;\approx\; 2.618034. \tag{6}$$

On substrates where $\inf \sigma(L_M) > 0$ (e.g. Dirichlet intervals), Eq. (5) gives the strict inequality $\|C_\varphi^{-1}\| < \varphi^2$. The response-amplification ceiling is, in either case, $\|\psi\|_2 \le \|C_\varphi^{-1}\| \, \|f\|_2$. On substrates with $\inf \sigma(L_M) > 0$ a finite ceiling is already guaranteed by the Dirichlet structure alone; on substrates with $\inf \sigma(L_M) = 0$ (the cases of interest here) the shift φ^{-2} is what guarantees a finite ceiling and pins it at φ^2. Numerically reproduced as $\|C_\varphi^{-1}\| = 2.618034$ on V_{600} (`repro/verify_kernel.py`, §4.1); this matches the closed-form φ^2 to machine precision.

2.5 Continuum projection

In one projected coordinate $x \in \mathbb{R}$ with $L_\varphi = -d^2/dx^2 + \varphi^{-2}$, the Green's function $G(x)$ satisfies $L_\varphi G = \delta_0$ and is the closed-form exponential

$$G(x) \;=\; \frac{\varphi}{2}\, e^{-|x|/\varphi}. \tag{7}$$

The decay scale is φ — the same constant that appears in the shift, by construction. Normalised, the kernel is $\kappa(x) = e^{-|x|/\varphi}$ with unit value at the source.

This continuum Green's function is the comparison object for the discrete-to-continuum agreement test (§5): the discrete response $\psi(v) = C_\varphi^{-1} f(v)$ at a vertex v at chord distance $\|v - v_{\mathrm{src}}\|$ from a localised source is compared to $G(\|v - v_{\mathrm{src}}\|)$.

2.6 Disclosure: φ^{-2} is a design-level shift

The shift φ^{-2} is chosen so that:

1. C_φ is strictly positive definite (the smallest eigenvalue is exactly φ^{-2});
2. φ^2 is the zero-bottom operator-norm *identity* ($\|C_\varphi^{-1}\| = \varphi^2$ when $\inf \sigma(L_M) = 0$) and the general response-amplification *ceiling* ($\|C_\varphi^{-1}\| \le \varphi^2$ on positive-bottom substrates), and the continuum decay scale is φ (Eq. (7)). Zero-bottom identity, general ceiling, and continuum decay scale are all fixed by the single design choice φ^{-2}, giving one dimensional parameter throughout the operator;
3. the continuum projection (Eq. (7)) has decay scale φ, not a free length parameter.

We do *not* derive φ^{-2} from a closure functional or symmetry argument. It is a design-level choice motivated by (1)–(3); we report this explicitly and treat formal derivation as an open build (§9). The companion adaptive-closure-transport preprint [Smart, 2026a] formulates the selection-layer dynamics over W-space that would, if delivered, constrain the shift further; that derivation is not delivered there or here.

3 The 600-cell substrate

This section gives the discrete instance used by the two empirical witnesses: $M = V_{600}$, the 600-cell regular 4-polytope under H_4 Coxeter symmetry, with the standard short-edge graph Laplacian. The choice of this polytope is post-hoc motivated by the empirical landings (§9); the construction itself is a classical Coxeter-group result we cite, not prove, here [Coxeter, 1973]. All facts in this section are reproduced numerically by `repro/verify_kernel.py` from the canonical generators alone.

3.1 Vertex set

V_{600} has $|V| = 120$ unit vectors on the unit 3-sphere $S^3 \subset \mathbb{R}^4$ [Coxeter, 1973, Weisstein]. With $\varphi = (1+\sqrt{5})/2$ the canonical vertex list partitions into three standard coordinate families:

- **Axis family** (8 vertices): all permutations of $(\pm 1, 0, 0, 0)$;
- **Half-integer family** (16 vertices): all sign combinations of $(\pm 1, \pm 1, \pm 1, \pm 1)/2$;
- **φ-mixed family** (96 vertices): all even permutations of $(\pm\varphi, \pm 1, \pm 1/\varphi, 0)/2$ (with the $\varphi^2 = \varphi + 1$ identity, equivalently $(\pm\varphi, \pm 1, \pm\varphi^{-1}, 0)/2$).

The total is $8+16+96 = 120$ unit vectors. These are coordinate families, not H_4 orbits: H_4 acts transitively on the full 120-vertex set, so the three families lie in a single H_4 orbit. Reproduced by `repro/verify_kernel.py:build_v600`; the numerical check $\max_v |\,\|v\| - 1\,| < 10^{-10}$ confirms all vertices on S^3.

The H_4 Coxeter group (order 14400) acts transitively on the 120 vertices. Every vertex therefore has *identical* local structure under H_4; in particular, every vertex has the same degree in the short-edge graph defined below.

3.2 Short-edge nearest-neighbour graph

For two unit vectors $v, w \in V_{600}$ on S^3, the Euclidean chord distance is

$$\|v - w\| \;=\; \sqrt{2 - 2\,\langle v, w\rangle}.$$

The *short-edge graph* $G_{V_{600}} = (V, E)$ connects two vertices if their inner product equals the canonical short-edge value

$$\langle v, w\rangle \;=\; \varphi/2 \;\approx\; 0.809, \tag{8}$$

equivalently chord distance $\|v - w\| = \sqrt{2-\varphi} = 1/\varphi \approx 0.618$. This is the nearest-neighbour adjacency on the canonical 600-cell embedding into S^3 [Coxeter, 1973].

Graph facts (forced by the construction). The graph $G_{V_{600}}$ has:

- $|V| = 120$ vertices,
- $|E| = 720$ edges,
- every vertex has degree exactly 12 (H_4 transitivity on the vertex set forces *uniformity* of the local structure; the short-edge nearest-neighbour construction gives the numerical degree 12, verified by `repro/verify_kernel.py`),
- the graph is connected (verified numerically by counting connected components of the short-edge adjacency matrix; the classical 600-cell connectivity result is well known in [Coxeter, 1973]).

All four facts are reproduced numerically: `repro/verify_kernel.py` reports $|V| = 120$, $|E| = 720$, degree-min/max = 12/12 (uniform), and one connected component.

3.3 9-shell H_3 partition

Choose any vertex v_0 as the pole; the H_3 subgroup of H_4 fixing v_0 partitions the remaining 119 vertices into shells of constant inner product with v_0. The nine canonical inner products are

$$\langle v, v_0 \rangle \in \{1,\ \varphi/2,\ 1/2,\ 1/(2\varphi),\ 0,\ -1/(2\varphi),\ -1/2,\ -\varphi/2,\ -1\}, \tag{9}$$

indexing shells $s = 0, 1, \ldots, 8$ from the pole to the antipode. The shell-size sequence is

$$(|S_0|, |S_1|, \ldots, |S_8|) = (1,\ 12,\ 20,\ 12,\ 30,\ 12,\ 20,\ 12,\ 1). \tag{10}$$

The middle shell S_4 has 30 equatorial vertices forming the icosidodecahedral ring. The total is $1+12+20+12+30+12+20+12+1 = 120$, matching $|V|$. Reproduced verbatim by `repro/verify_kernel.py:shell_indices`.

Antipodal symmetry. The map $v \mapsto -v$ takes the shell-s vertices to the shell-$(8-s)$ vertices: $s(-v) = 8 - s(v)$. The antipode $-v_0$ is the unique shell-8 vertex.

3.4 Inner-product check

The canonical short-edge criterion (Eq. (8)) and the canonical shell inner products (Eq. (9)) are jointly consistent: a vertex in shell s_1 is connected to a vertex in shell s_2 if and only if their pairwise inner product is $\varphi/2$, which restricts the admissible (s_1, s_2) adjacency types to those compatible with the H_3 orbit structure. The numerically constructed graph respects this: every edge has inner product exactly $\varphi/2$ within machine precision (tolerance 10^{-10} in `repro/verify_kernel.py:build_short_edge_graph`).

3.5 What the substrate fixes, and what it does not

- **Fixed by the construction once V_{600} is chosen**: $|V| = 120$, uniform degree 12, 9-shell partition $\{1, 12, 20, 12, 30, 12, 20, 12, 1\}$, antipodal symmetry, and the Laplacian spectrum (§4).
- **Fixed by the design-level φ^{-2} shift**: C_φ is positive definite with smallest eigenvalue φ^{-2} (§2.4); the operator-norm bound $\|C_\varphi^{-1}\| = \varphi^2$.
- **Not fixed by this paper**: the choice of V_{600} over the 24-cell, 120-cell, or other regular 4-polytopes / Coxeter substrates. The 600-cell choice is post-hoc motivated by the empirical landings (§6, §7). A formal substrate ablation is an open build (§9).

4 The Laplacian spectrum

The unweighted graph Laplacian $L_{V_{600}} = D - A$ on V_{600} numerically resolves into nine distinct eigenvalue classes whose values match the closed-form $\mathbb{Z}[\varphi]$ list given in Table 1 to machine precision; multiplicities sum to $|V| = 120$. The spectrum is computed numerically by `repro/verify_kernel.py:laplacian_spectrum` (a single 120×120 symmetric eigendecomposition, deterministic at machine precision); the closed-form identification is made by algebraic recognition of the displayed values, not by an exact arithmetic derivation in this paper.

Table 1: Computed Laplacian spectrum of $L_{V_{600}}$ on V_{600}. Closed-form values in $\mathbb{Z}[\varphi]$ alongside the numerical values returned by `repro/verify_kernel.py`; multiplicities sum to 120.

Closed-form eigenvalue	Numerical value	Multiplicity
0	machine zero ($\sim 10^{-15}$)	1
$12 - 6\varphi$	2.2918	4
$12 - 4\varphi$	5.5279	9
9	9.0000	16
12	12.0000	25
14	14.0000	36
$4\varphi + 8$	14.4721	9
15	15.0000	16
$6\varphi + 6$	15.7082	4
	Total multiplicity:	**120**

Closed-form check. Using $\varphi = (1 + \sqrt{5})/2$:

$$\begin{aligned}
12 - 6\varphi &= 12 - 3(1 + \sqrt{5}) = 9 - 3\sqrt{5} \approx 2.2918, \\
12 - 4\varphi &= 12 - 2(1 + \sqrt{5}) = 10 - 2\sqrt{5} \approx 5.5279, \\
4\varphi + 8 &= 2(1 + \sqrt{5}) + 8 = 10 + 2\sqrt{5} \approx 14.4721, \\
6\varphi + 6 &= 3(1 + \sqrt{5}) + 6 = 9 + 3\sqrt{5} \approx 15.7082.
\end{aligned}$$

The eigenvalue pairs $\{12-6\varphi,\ 6\varphi+6\}$ (both with multiplicity 4) and $\{12-4\varphi,\ 4\varphi+8\}$ (both with multiplicity 9) are conjugate under the Galois automorphism $\sigma\colon \sqrt{5} \mapsto -\sqrt{5}$ on $\mathbb{Z}[\varphi]$. The fixed-point eigenvalues $\{0, 9, 12, 14, 15\}$ are rational and have multiplicities $\{1, 16, 25, 36, 16\}$ (sum 94); the σ-paired eigenvalues have total multiplicity $4 + 4 + 9 + 9 = 26$.

σ-fix vs σ-paired multiplicity split. $94/120 = 78.3\%$ of the spectrum is σ-fixed (rational); the remaining $26/120 = 21.7\%$ is σ-paired. We report this structural fact about the spectrum and do not use it in any claim made by this paper.

4.1 Operator-norm bound on C_φ

The smallest eigenvalue of $C_\varphi = L_{V_{600}} + \varphi^{-2} I$ is

$$\lambda_{\min}(C_\varphi) \;=\; 0 + \varphi^{-2} \;=\; \varphi^{-2} \;\approx\; 0.381966,$$

and the operator-norm bound is

$$\|C_\varphi^{-1}\| \;=\; 1/\varphi^{-2} \;=\; \varphi^2 \;\approx\; 2.618034.$$

`repro/verify_kernel.py:operator_norm_check` reports $\|C_\varphi^{-1}\| = 2.618034$ (numerical) vs $\varphi^2 = 2.618034$ (predicted) — match to six decimal places. The largest eigenvalue of C_φ is $\lambda_{\max}(L_{V_{600}}) + \varphi^{-2} = (6\varphi + 6) + \varphi^{-2} = 9 + 3\sqrt{5} + (2 - \varphi) \approx 16.0902$.

4.2 H_4 irrep block decomposition (imported context)

The eigenspaces of $L_{V_{600}}$ partition into H_4-proper and σ-twin Coxeter exponent classes. For H_4 proper the exponents are $\{1, 11, 19, 29\}$; under the σ-automorphism of $\mathbb{Z}[\varphi]$ the exponents become $\{7, 13, 17, 23\}$. The σ-orbit projector basis used in the aria-chess companion's recurrent layer [Smart, 2026b] realises this block decomposition at machine precision and provides a spectrally clean H_4-equivariant basis there.

This subsection is imported context from the aria-chess companion; the irrep block decomposition is *not* verified by `repro/verify_kernel.py` of this paper and is not used as a load-bearing fact for any operator-witness claim made here. We include the labelling for orientation only.

5 Discrete-to-continuum agreement

This is the central geometric diagnostic of the paper: the discrete response $\psi = C_\varphi^{-1} f$ on V_{600} for a localised source has high per-vertex Pearson correlation in radial shape with the continuum kernel $G(x) = (\varphi/2)\, e^{-|x|/\varphi}$ at the vertex's chord distance from the source. Pearson correlation is a shape similarity statistic, not an equality claim. We give the test, the numerical result, and a variant comparison in which the unweighted Laplacian ranks highest among the three tested variants (unweighted, φ-geometric weighted, φ-arithmetic weighted).

5.1 The test

Pick a pole vertex v_0 (we use the canonical $+x_0$ axis vertex). Set $f = e_{v_0}$ (the unit vector at v_0, all other entries zero). Compute

$$\psi \;=\; C_\varphi^{-1} f \;=\; (L_{V_{600}} + \varphi^{-2} I)^{-1} e_{v_0}$$

by direct linear solve (no eigenmode truncation). For each vertex $v \in V$, compute the Euclidean chord distance $x(v) = \|v - v_0\|$ and the continuum prediction

$$G(x(v)) \;=\; (\varphi/2)\, \exp(-\, x(v)/\varphi).$$

The agreement criterion is the Pearson correlation between $\psi(v)$ and $G(x(v))$ across $v \in V \setminus \{v_0\}$ (the source itself is excluded, since the discrete response there is trivially the diagonal of C_φ^{-1} and the chord distance is zero, both degenerate for the comparison).

5.2 Result on the unweighted Laplacian

`repro/verify_kernel.py:variant_correlation` returns:

- **Multiplicity-weighted per-vertex Pearson correlation over shell-constant responses**: $\rho = 0.976$. (ψ is shell-constant to machine precision under H_3-fixed source, so the per-vertex test is mathematically a multiplicity-weighted shell-radius correlation rather than 119 independent radial samples; see the within-shell-variance discussion at the end of this subsection.)
- **Shell-mean Pearson correlation**: $\rho = 0.923$ (averaging $\psi(v)$ over each H_3 shell first, then correlating the 9-point shell-mean trajectory with the continuum prediction at the shell mean radius).

The two correlations measure the same shell-radius fact at different weightings and with different source-vertex conventions:

- Per-vertex test: $|V| - 1 = 119$ data points (every non-source vertex), source v_0 *excluded* (the discrete response there is the diagonal of C_φ^{-1} and the chord distance is 0, both degenerate for the comparison).

- Shell-mean test: 9 data points (one per H_3 shell); shell 0 contains only the source vertex, so it is included on the shell-mean side and contributes a single $(\psi(v_0), G(0))$ point.

On the unweighted 600-cell graph with an H_3-fixed source, ψ is shell-constant up to numerical precision — the within-shell standard deviations are at machine precision ($\sim 10^{-16}$). The two tests therefore differ in weighting and source convention, not in noise content: the per-vertex test weights each shell by its multiplicity ($\{12, 20, 12, 30, 12, 20, 12, 1\}$ for the non-source shells) and excludes the source vertex, while the shell-mean test gives equal weight to every shell and includes the source. The per-vertex test is the headline agreement criterion in this paper.

5.3 Variant comparison

Two φ-cocycle weighted Laplacian variants are tested as controls:

- **φ-geometric weights**: edge weight $w_{vw} = \sqrt{\omega_+(v)\,\omega_+(w)}$ with vertex weight $\omega_+(v) = \varphi^{\kappa(v)}$, where $\kappa(v) \in \{0, \ldots, 8\}$ is the shell index of v.
- **φ-arithmetic weights**: edge weight $w_{vw} = \frac{1}{2}[\omega_+(v) + \omega_+(w)]$ with the same ω_+.

The weighted Laplacian is then $L_w = D_w - A_w$ where A_w is the weighted adjacency. We re-run the discrete-to-continuum test on each variant.

Table 2: Per-vertex and shell-mean Pearson correlations between the discrete response $\psi = (L_w + \varphi^{-2}I)^{-1}e_{v_0}$ and the continuum prediction $G(\|v - v_0\|)$ for three Laplacian variants (L_w unweighted or φ-cocycle weighted). Computed by `repro/verify_kernel.py:variant_correlation`.

Variant	Per-vertex correlation	Shell-mean correlation
Unweighted	**0.976**	**0.923**
φ-geometric weighted	0.888	0.880
φ-arithmetic weighted	0.884	0.878

Reading. Among the three tested variants, the unweighted Laplacian ranks highest on both reported criteria (+0.088 per-vertex over the next variant, +0.044 shell-mean). This reproduces, on a different test, the qualitative ranking the b-anomaly paper [Smart, 2026c] established independently against its data-χ^2 criterion on the LHCb 2025 dataset (see §6 for the b-anomaly numbers). Two independent criteria — geometry-only correlation here, and angular-anomaly χ^2 in b-anomaly — agree on which of the three tested variants ranks highest. We do not claim this is a uniqueness or blind-selection result; we report it as a two-criterion agreement on the highest-ranked tested variant (the b-anomaly paper's own caveat that the data was looked at first and the geometry criterion verified afterward is inherited verbatim).

5.4 What the agreement does and does not establish

Does establish. A geometric agreement: the discrete response of a fixed-shift Green operator on a fixed graph correlates per-vertex in radial shape, at Pearson 0.976, with the closed-form continuum exponential at the same length scale φ. This is a non-trivial Pearson correlation between separately evaluated discrete and continuum Green responses sharing the same design-level scale φ^{-2}: (i) the discrete inverse of a 120×120 Laplacian-plus-shift matrix; and (ii) a one-dimensional continuum exponential. The φ-mediated agreement is an empirical fact about the chosen substrate and shift, computed (not fitted) by the verification script.

Does not establish. Operator uniqueness on either empirical landing — the b-anomaly paper documents a free-width Gaussian alternative that fits comparably in χ^2 at the cost of one extra shape parameter, and the aria-chess preprint does not run a substrate ablation; both caveats are inherited verbatim. The agreement also does not establish that V_{600} is the unique discrete substrate with this property; the 24-cell, 120-cell, and other H_n Coxeter graphs would need to be tested on the same correlation criterion to make any comparative claim, and a formal substrate ablation is an open build (§9).

6 Passive-regime witness: b-anomaly

This section threads the first of the two domain-disjoint empirical landings of C_φ. The full preprint is [Smart, 2026c]; we summarise here only what the operator-witness narrative requires and inherit the preprint's caveats verbatim.

6.1 What b-anomaly tests

The Wilson-coefficient Hamiltonian for $b \to s\mu^+\mu^-$ contains a $C_9^{(\prime)}$ contribution that, in the Standard Model, is approximately q^2-independent in the relevant kinematic range. The b-anomaly preprint tests a fixed q^2-dependent effective shift on top of the SM backend, of the form

$$\Delta C_9^{\mathrm{eff}}(q^2) \;=\; -\,A \cdot \kappa_{V_{600}}(q^2), \tag{11}$$

where $\kappa_{V_{600}}(q^2) > 0$ on the LHCb q^2 window is the projection of C_φ on V_{600} to the flavour-physics q^2 axis (the b-anomaly preprint's §3 projection construction, which we do not relitigate here; this is a projection of the operator, not a derivation of φ^{-2}), and A is a single fitted dimensionless amplitude per dataset. The explicit minus sign in Eq. (11) reflects the b-anomaly preprint's sign convention: a positive fitted $A > 0$ produces a negative $\Delta C_9^{\mathrm{eff}} < 0$, the established direction of the angular anomaly. The kernel shape $\kappa_{V_{600}}$ is held fixed across all five datasets. This is a *structural* test: same fixed C_φ on the same V_{600}, no shape retuning between datasets.

6.2 The five-dataset structural fit

The b-anomaly preprint reports the following per-dataset table (verbatim from [Smart, 2026c], also at `BANOMALY-001/vfd-b-anomaly/README.md`):

Table 3: b-anomaly five-dataset structural fit. Verbatim from [Smart, 2026c]; one fitted amplitude A per dataset, kernel shape held fixed.

Dataset	Decay	n	$\Delta\mathrm{AIC}_{\mathrm{NL}}$	Best-fit A	$\Delta C_9^{\mathrm{eff}}$
LHCb 2015	$B^0 \to K^{*0}$	32	-0.24	$+1.24$	-0.96
LHCb 2021	$B^+ \to K^{*+}$	32	$+0.17$	$+2.06$	-1.59
CMS 2025 (no P_4')	$B^0 \to K^{*0}$	18	$+0.47$	$+1.05$	-0.81
LHCb 2025	$B^0 \to K^{*0}$	32	$+1.09$	$+1.14$	-0.86
LHCb 2015	$B_s \to \phi$ (S-basis)	24	-0.24	$+4.98$	-3.85

6.3 What the structural fit reports

- **Cross-dataset consistency (5/5).** The same fixed kernel shape can be fit to all five datasets with one amplitude A per dataset and no shape retuning. The kernel never moves between datasets.

- **Sign uniformity (5/5).** $A > 0$ in 5/5 fits; $\Delta C_9^{\text{eff}} < 0$ in 5/5 fits. The kernel reproduces the established direction of the anomaly [LHCb Collaboration, 2020] across all five independent measurements.

- **Cross-channel ratio.** The b-anomaly README reports the basis amplification (the predicted Krüger–Matias P-basis vs S-basis factor ~ 2.2 [Krüger and Matias, 2005]) as partly explanatory for the $B \to K^*$ vs $B_s \to \phi$ amplitude gap. Working through the explicit arithmetic with $B \to K^*$ best-fit $A_P \approx 1.14$: the basis-corrected prediction for $B_s \to \phi$ is $A_P \cdot 2.2 \approx 2.5$; the observed $B_s \to \phi$ amplitude is $A \approx 4.98$, leaving an unresolved amplitude excess of about a factor two above the basis-corrected prediction. The b-anomaly preprint reports this residual as an open item, not a discharge.

- **Geometry-first variant test.** Of three discrete Laplacian variants on V_{600} (unweighted, φ-geometric weighted, φ-arithmetic weighted), the unweighted choice ranks highest on both a pure-geometry criterion (correlation 0.997 with the continuum kernel under the b-anomaly preprint's q^2/shell-projection geometry metric, §3.4 of [Smart, 2026c] — *not* this paper's per-vertex test, whose unweighted score is 0.976; the two metrics differ in what is correlated) and the LHCb 2025 data χ^2 ($\chi^2 = 13.555$). The two criteria agree on the variant ranking — a two-criterion agreement on the same fixed operator.

6.4 What the structural fit does *not* establish

The b-anomaly preprint is explicit about the following caveats, which we inherit verbatim:

- **AIC tie on current data.** On Akaike model comparison, C_φ-derived $\kappa_{V_{600}}$ and a constant Wilson-coefficient shift (FREE_C9, also $k = 1$) are statistically indistinguishable: stacked AIC weights $w_{\text{VFD}} = 0.348$ vs $w_{\text{FREE\_C9}} = 0.652$. Current data cannot resolve the model comparison. AIC is blind to the universality / shape-prediction claim itself, but it is decisive about whether the shape is forced by data: it is not.

- **Free-width Gaussian alternative.** A free-width Gaussian charm-loop proxy fits the same five datasets comparably in χ^2 at the cost of one extra shape parameter; C_φ is not the unique q^2 shape consistent with the anomaly.

- **Mode-B drift (linearised-to-non-linear refit).** An earlier analysis the b-anomaly project labels Mode-B (linearised) gave a stronger preference ($\Delta\text{AIC} = -1.67$ on LHCb 2025) that did not survive the subsequent non-linear refit; the $+2.77$-AIC-unit drift between Mode-B (linearised) and the non-linear refit is the largest single methodological uncertainty in the b-anomaly project.

- **Look-elsewhere on the variant test.** The b-anomaly preprint's limitations section [Smart, 2026c] acknowledges that the LHCb 2025 data was looked at first, and only later was the agreement of the data-χ^2 ranking with the pure-geometry ranking verified. (The b-anomaly README emphasises that the geometry-only criterion is independent of the LHCb data; the historical non-blindness is in the project's ordering, not in the criterion.) The two-criterion agreement is criterion-independent but not historically blind.

6.5 Reading at the operator level

The b-anomaly result is the *passive-regime* empirical witness for C_φ on V_{600}: a single linear response $\psi = C_\varphi^{-1} f$, projected to the q^2 axis through a fixed discrete-to-momentum projection, gives a sign-uniform structural fit consistent with the $b \to s\mu^+\mu^-$ angular anomaly across five independent measurements without shape retuning. This does not establish the kernel as theorem-grade physics on the flavour side (the AIC tie, the free-width Gaussian alternative, and the Mode-B linearised-to-non-linear refit drift prevent that). It does support, at the operator-witness level, the inherited reading that the same fixed C_φ on the same fixed V_{600} is consistent with one of two domain-disjoint empirical landings without parameter retuning. The second landing is in §7.

7 Active-regime witness: aria-chess

This section threads the second of the two domain-disjoint empirical landings of C_φ. The full preprint is [Smart, 2026b]; we summarise here only what the operator-witness narrative requires and inherit the preprint's substrate-witness scope verbatim.

7.1 What aria-chess tests

The aria-chess preprint adds a recurrent self-model layer above the same C_φ on the same V_{600}. The architecture introduces:

- One *condition-dependent* self-injection coupling $\eta \in \{0, 0.05, 0.20\}$ (PROPOFOL, SLEEP_N3, WAKE/RECOVERY) that controls the strength of the recurrent feedback;
- One *substrate-pinned* nonlinearity bounded_topk$(\cdot, k = 12)$ at the graph's average degree (§3.2: degree 12 uniform). The choice $k = 12$ is not a free hyperparameter; it is the substrate's average degree.
- Condition-specific *biologically-motivated* stimulus models (slow oscillation + spindles + K-complexes for SLEEP_N3, AR(1) noise + tonic shell + attention episodes for WAKE, low-amplitude tonic noise for PROPOFOL). These are biologically-motivated design choices, not measurement-fits to subject-level EEG data.

The kernel parameter φ^{-2} is *not retuned* between b-anomaly and aria-chess; the same fixed shift used in the flavour-physics structural fit is used in the cortical substrate witness.

7.2 Eighteen preregistered correspondences

Eighteen quantitative predictions (P1–P18) were locked on 2026-04-18 in the aria-chess preprint's `docs/brain_mapping/PAPER_PREDICTIONS.md` before any validation run. Each prediction has a specific numerical claim, a falsifiable threshold, and a named validation script. The preregistered tally as reported in [Smart, 2026b]:

- 17/18 at standard validation methodology (5-seed cascade block plus state-reset protocol);
- 18/18 after a documented $N = 20$ deep-dive on the residual high-variance interaction $C \times P$ (P4: bootstrap point estimate +0.190, 95% CI $[+0.143, +0.239]$, 0/2000 resamples at-or-below zero, reported as 0.0000).
- No preregistered threshold has been modified.

The aria-chess preprint reports this as methodology refinement (documented seed-count increase on a high-per-seed-variance interaction term), *not* as a threshold change. Two further predictions in the aria-chess set carry their own documented methodology refinements that we also inherit: P3 ($D \times C$ independence) closes at $N = 5$ within its preregistered band, and P13 (chess substrate lift) was preregistered with 5-fold cross-validation but the landed value +40.6pp uses the documented leave-one-out / state-reset protocol; both refinements are reported in [Smart, 2026b] with no threshold modification, and we inherit those caveats verbatim. The aria-chess preprint also flags that the 2026-04-29 P10 (chess null mapping) validation used 15 permutations rather than the preregistered 20 (threshold $\geq 50\%$ unchanged; observed 65.4% robust in the 15–20 range; the prereg-exact 20-permutation rerun is left open in the aria-chess limitations); we inherit this caveat verbatim alongside the 18/18 summary.

7.3 Six drug/sleep EEG signatures

On a single deterministic substrate trajectory at seed 42, the aria-chess preprint reports six biological signatures testing against literature-derived thresholds:

- **Wake cortical-avalanche α**: $\alpha = 2.252$, 95% CI $[1.82, 2.86]$, $R^2 = 0.956$ — the WAKE confidence interval overlaps both the Sleep-EDFx EEG CI $[2.50, 2.53]$ ($n = 30$ subjects) and aria-chess's prior cascade pipeline CI $[2.73, 3.25]$ pairwise (the Sleep-EDFx and prior-pipeline intervals do not overlap each other; the WAKE interval is the pairwise common ground).
- **NREM-N3 phenomenal-intensity variance ratio**: $0.463\times$ wake (predicted ~ 0.365, threshold < 0.70).
- **Propofol modality-switching ratio**: $1.83\times$ wake (threshold $\in [1.5, 5.0]$, empirical reference $2.96\times$ from OpenNeuro `ds005620`).
- **Propofol continuity drop**: $+0.066$ (threshold > 0.020).
- **Propofol Φ collapse**: $0.33\times$ wake (IIT direction confirmed; Φ-proxy not full IIT).
- **Recovery deterministic identity to wake**: under the WAKE stimulus protocol, the RECOVERY trajectory is bit-identical to the WAKE trajectory.

7.4 Cross-domain selectivity

- **Chess pattern recognition (P9–P13)**: 32 chess positions across 4 categories on 8-D V2 features; under the disclosed leave-one-out / state-reset refinement of the preregistered $\geq$ +15pp substrate-lift test (the original prereg used 5-fold cross-validation; the LOO/state-reset protocol is documented in the aria-chess preprint [Smart, 2026b]), substrate routing lifts classification at canonical depth $n = 25$ ticks from raw 53.1% to substrate-routed 93.8% (+40.6pp), above the preregistered $\geq$ +15pp floor.
- **Conversation pattern recognition (P14–P16)**: 64 utterances, 8 categories; raw classification 87.5%, substrate-routed lift -4.4pp (within the preregistered neutrality band $|\cdot| < 10$pp). The substrate is selectively amplifying in tasks where raw features are ambiguous and approximately neutral when raw features are already discriminative.
- **HCP brain functional connectivity (P17–P18)**: full-cohort descriptive statistics on $n = 1003$ subjects show ARIA's H_4-transitive structure at -11.58σ on degree homogeneity, $+79.78\sigma$ on raw participation ratio (with node-count caveat: ARIA $|V| = 120$ vs HCP ICA-50 $|V| = 50$), and $+6.80\sigma$ on clustering coefficient. ARIA's degree std is 0 by H_4 transitivity (a theorem), 11.58 standard deviations below the HCP biological cohort.

7.5 Reading at the operator level

The aria-chess result is the *active-regime* empirical witness for C_φ on V_{600}. The recurrent self-model layer above C_φ uses one condition-dependent coupling and one substrate-pinned nonlinearity at the graph's average degree $k = 12$; no other *kernel-shape* parameter enters. Above the operator, aria-chess inherits its own dynamical and stimulus constants (e.g. a fixed dynamical decay, fixed cascade gains, condition-specific biologically-motivated stimulus models); these are documented in the aria-chess preprint and are not retuned in this paper. The kernel shift φ^{-2} is not retuned between b-anomaly and aria-chess. Under those design choices, the same fixed C_φ on the same V_{600} is consistent with the aria-chess active-regime tally (frozen 2026-04-18): 17/18 preregistered cortical correspondences under the standard validation protocol, 18/18 after the documented $N = 20$ P4 deep-dive (thresholds unchanged), plus six literature-thresholded EEG drug/sleep signatures.

The aria-chess preprint stays inside substrate-witness scope: it does not claim the substrate *is* consciousness, does not claim 600-cell uniqueness among regular 4-polytopes, and does not deliver a selection theorem on the ACT 4-tuple. We inherit the scope verbatim. What we add at the operator level is the observation that the same fixed C_φ — under no shape-parameter retuning between regimes — is the shared response primitive underneath both empirical landings (the b-anomaly q^2 projection above C_φ and the aria-chess recurrent self-model layer above C_φ are distinct above-operator constructions; the operator below them is the same).

7.6 Two-witness structure

Table 4: Two domain-disjoint empirical landings of C_φ on V_{600}, with no shape retuning between regimes.

	Passive regime	Active regime
Preprint	b-anomaly [Smart, 2026c]	aria-chess [Smart, 2026b]
Domain	flavour physics	cortical neuroscience
Datasets	5 public ($n_{\text{tot}} = 138$ bins)	prereg + EEG (HCP $n = 1003$, Sleep-EDFx $n = 30$, etc.)
Operator	same fixed C_φ	same fixed C_φ
Substrate	same V_{600}	same V_{600}
Shift φ^{-2}	not retuned	not retuned
Headline	5/5 sign uniformity	17/18 standard, 18/18 after P4 $N = 20$ deep-dive (P3 $N = 5$ + P13 LOO/state-reset documented refinements; P10 prereg-exact 20-perm rerun open); 6/6 EEG
Kernel-shape free params	0	0
Other free params	1 amplitude A per dataset	1 coupling η per condition; v4 stimulus protocols
Caveat	AIC tie; free-width Gaussian alt.; Mode-B linearised-to-non-linear refit drift	single-seed EEG; no polytope ablation; P10 prereg-exact rerun open

The two witnesses share, by design, exactly the geometry-fixed operator: the same C_φ, the same substrate V_{600}, and the same shift φ^{-2}. They share no fitted parameter, threshold, dataset, or methodological choice above the operator level. The b-anomaly amplitude A is fitted to flavour-physics q^2 shape, dataset by dataset; the aria-chess coupling η is fixed by experimental condition (PROPOFOL/SLEEP/WAKE), not by neural data; the v4 stimulus protocols are biologically motivated, not subject-fit. Independence in this paper is independence at the empirical layer above the shared operator.

8 Programme home and the open selection layer

This section positions C_φ within the broader cascade programme and identifies what the operator does not deliver. The framing matters for the hostile-review reading: C_φ is the *response* primitive; *selection* (which response configuration the system picks under dynamics) is a separate layer that this paper does not close.

8.1 Programme position

C_φ is one instance of a polynomial-in-L functional on a finite-dimensional substrate. The published companion adaptive- closure-transport preprint [Smart, 2026a] positions a homeostatic regulariser R_{hom} in the 4-tuple $(M, L_M, W, R_{\text{hom}})$ as a member of the same polynomial-in-L family. We use this as the only published programme-level reference point for C_φ. We make no classification claim, no family-membership theorem, and no forecast about any other members of this programme; any broader programme arc lies outside the operator-witness scope of this paper.

8.2 Response vs selection

The closure response $\psi = C_\varphi^{-1} f$ is determined by the chosen substrate plus the design-level shift: C_φ is fixed by the substrate V_{600} and the design-level choice φ^{-2}, and the response is the resulting linear inverse. This is a *response* primitive. It does *not* answer:

- Why this substrate? (Selection across regular 4-polytopes $\{\text{24-cell}, \text{600-cell}, \text{120-cell}\}$.)
- Why this shift? (Selection of φ^{-2} over an arbitrary positive constant.)
- How does the system pick a response configuration over time? (Crystallisation / Lyapunov descent dynamics on a W-trajectory.)

The selection layer is open. The companion adaptive-closure- transport preprint [Smart, 2026a] proposes a Lyapunov $V(W)$ on the reduced flow, an edge-space decomposition under the $2I$ binary-icosahedral symmetry of V_{600}, and a full reduced-flow convergence theorem on W-trajectories; that paper explicitly does *not* deliver those, and we do not deliver them here. The aria-chess companion [Smart, 2026b] likewise stays inside substrate-witness scope and does *not* deliver a selection theorem. The two empirical witnesses landed in this paper provide external consistency checks on the *response* primitive without reducing or addressing the selection gap.

8.3 What this paper closes vs leaves open

Closes (at the operator-witness level).

- The operator C_φ is well-defined and positive definite on any (M, L_M) satisfying (H1)–(H3); the operator-norm identity $\|C_\varphi^{-1}\| = \varphi^2$ holds whenever $\inf \sigma(L_M) = 0$ (e.g. on a connected finite graph with the standard combinatorial Laplacian, or on the full-line continuum case). On substrates where $\inf \sigma(L_M) > 0$ (e.g. Dirichlet-boundary continuum cases) the bound $\|C_\varphi^{-1}\| \leq \varphi^2$ holds and is generally strict (§2).
- The 600-cell instance V_{600} has the construction described (§3) and the Laplacian spectrum of Table 1, both reproduced numerically (`repro/verify_kernel.py`).
- Discrete-to-continuum agreement at per-vertex Pearson correlation 0.976 on the unweighted variant, with the unweighted variant winning the geometry-only criterion against two φ-cocycle weighted controls (§5).
- Same fixed C_φ on same fixed V_{600} appears as the shared response primitive underneath two domain-disjoint empirical works in qualitatively distinct regimes (§6, §7).

Leaves open.

- *First-principles derivation of φ^{-2}.* Reported as a design-level shift; not derived from a closure functional or symmetry argument.
- *Substrate-uniqueness ablation.* The 600-cell choice is post-hoc motivated by the empirical landings; alternative regular 4-polytopes are an explicit ablation build, not a discharged comparison.

- *Kernel-uniqueness on either empirical landing.* The b-anomaly free-width Gaussian alternative (fits comparably with one extra shape parameter), the AIC tie ($w_{\mathrm{VFD}} = 0.348$ vs $w_{\mathrm{FREE\_C9}} = 0.652$), and the Mode-B linearised-to-non-linear refit drift caveat are inherited verbatim from [Smart, 2026c].
- *Selection theorem on ACT.* Lyapunov $V(W)$, edge-space decomposition under $2I$-equivariance, full reduced-flow convergence — all explicitly not delivered in [Smart, 2026a] and not delivered here.
- *Family-membership theorem.* The programme-home positioning of cascade Lyapunov functionals as members of the same polynomial-in-L family is reported as *programme-positioned*, not formally classified.

9 Limitations and hostile-review guard matrix

This section enumerates limitations transparently, organised as a five-move guard matrix following the b-anomaly preprint template [Smart, 2026c]: regime, post-hoc, interpretation, test/claim, state-drift. For each guard we record $G\colon$ risk $\to$ (disclosure, evidence, strengthening build).

9.1 Regime

Single substrate (the 600-cell). We have not tested whether C_φ on the 24-cell, the 120-cell, or other H_n Coxeter graphs would give comparable per-vertex correlations on the discrete-to-continuum agreement test, or comparable structural fits on either empirical landing. The 600-cell choice is post-hoc motivated by the empirical landings, not from an a-priori derivation. *Disclosure:* §1, §3, §8. *Evidence:* per-vertex correlation 0.976 on V_{600}; empirical landings of §6 and §7. *Strengthening build:* `repro/verify_kernel.py` extension to the 24-cell and 120-cell, with the same per-vertex correlation criterion applied; verbatim re-run of the b-anomaly fit on alternative substrates from [Smart, 2026c]; the aria-chess preprint's regime section already lists alternative-polytope ablation as an open build.

Single shift (φ^{-2}). We have not tested whether nearby shifts ($\varphi^{-2} \pm \epsilon$ for small ϵ) give comparable per-vertex correlation, or whether the empirical landings tolerate a shift sweep. The shift is held fixed across both regimes; perturbation analysis is an open build. *Strengthening build:* sweep $\varphi^{-2} \cdot (1 \pm \delta)$ for $\delta \in \{0.01, 0.05, 0.10, 0.25\}$ on the discrete-to- continuum correlation; report sensitivity envelope.

9.2 Post-hoc

The 600-cell choice is post-hoc justified by empirical observables. While the construction of V_{600} is theorem-level rigorous (H_4 Coxeter group theory), the choice of *this* polytope as the discrete substrate instance is motivated by the empirical landings observed, not by an a-priori geometric or algebraic argument selecting it over the 24-cell or 120-cell. *Disclosure:* §1. *Evidence:* two domain-disjoint empirical witnesses on V_{600}. *Strengthening build:* formal substrate ablation (above).

The geometry-first variant agreement is not historically blind on b-anomaly. Per the b-anomaly preprint's limitations section, the LHCb 2025 data was inspected first and the pure-geometry ranking was verified afterward to agree with the data-χ^2 ranking. The two-criterion agreement is *criterion-independent* (geometry-only correlation here is a different test from b-anomaly's data χ^2) but not historically pre-registered. *Disclosure:* we inherit the caveat verbatim. *Strengthening build:* a future blind variant test would freeze the variant choice before observing the data χ^2.

φ^{-2} floor not derived. The shifted-Laplacian floor $\varphi^{-2} \approx 0.382$ is a stability clamp making C_φ strictly positive definite; it is not derived from a closure functional or symmetry argument.

Disclosure: §2.4, §2. *Evidence:* the same operator with the same shift serves as the basis for two domain-disjoint empirical witnesses across qualitatively distinct regimes (§6, §7). *Strengthening build:* derive or justify a distinguished shift under a named criterion (e.g. minimum-amplitude amplification on a specified function class); uniqueness is not assumed and is itself an open problem.

9.3 Interpretation

The discrete-to-continuum agreement is descriptive, not causal. The per-vertex correlation 0.976 between ψ on V_{600} and the continuum kernel $G(x) = (\varphi/2)\, e^{-|x|/\varphi}$ at the same chord radii is a *computed agreement* between two independently-defined objects, not a derivation that the discrete operator equals the continuum kernel. *Disclosure:* §5 marks this explicitly. *Evidence:* the agreement is at machine precision in the operator-norm bound and at $\rho = 0.976$ in the per-vertex correlation. *Strengthening build:* a formal discrete-continuum convergence proof under a specified large-graph limit; a continuum limit theorem on H_n Coxeter substrates as $n \to \infty$.

Variant ranking is criterion-dependent. The unweighted variant wins on both the geometry-only criterion of this paper and b-anomaly's data χ^2, but neither criterion is the *unique* natural ranking. Edge-weighted variants outside the φ-cocycle family ($\sqrt{\text{deg}}$-weighted, normalised Laplacian) are not tested here. *Strengthening build:* extend `repro/verify_kernel.py` to a wider variant catalogue.

9.4 Test/claim

Two domain-disjoint empirical landings, not formal physics. The b-anomaly result is a structural fit (kernel shape held fixed) under an AIC tie with FREE_C9 on current data; the aria-chess result is a substrate witness (no claim that the substrate *is* consciousness). Neither lands a theorem-grade physics claim on its own domain; both are appropriately hedged in their own preprints, and we inherit those hedges verbatim. *Disclosure:* §6, §7. *Evidence:* the witnesses pass their own preregistered or literature-derived thresholds. *Strengthening build:* more flavour-physics datasets (LHCb Run 3, Belle II) for the passive-regime witness; cross-cohort EEG (TUH, NHM) and cross-parcellation HCP (Schaefer, Glasser) for the active-regime witness; both already listed in the respective preprints.

Discrete-to-continuum correlation reported on a single pole, verified across all $|V|$ on the unweighted operator. The headline per-vertex correlation 0.976 is reported with the canonical pole ($+x_0$ axis) as source on the unweighted Laplacian. H_4 transitivity predicts the correlation is invariant under choice of source vertex. The `multi_source_sweep` routine in `repro/verify_kernel.py` runs over all 120 vertices on the unweighted operator only (the two φ-cocycle weighted variants are not swept here, an explicit ablation build); every unweighted source returns the same per-vertex correlation 0.976202 to within $\sim 10^{-15}$ (`multi_source_sweep.max_minus_min` in `results.json`). The single-pole headline is therefore a representative, not a sample, of the unweighted operator's behaviour under H_4 transitivity; the two φ-cocycle weighted variants are not covered by this sweep.

9.5 State-drift / out-of-scope

Selection layer not delivered. As enumerated in §8, the selection-layer constructions (Lyapunov $V(W)$, edge-space decomposition under $2I$-equivariance, full reduced-flow convergence) are open in the companion ACT preprint [Smart, 2026a] and are not delivered here. The two empirical witnesses strengthen confidence in the response primitive without addressing the selection gap.

Broader cascade programme is out of scope. Other preprints in the broader cascade programme share operator-level infrastructure with this paper but are not load-bearing here. Deliberately out of scope.

Out-of-scope items NOT delivered (correctly).

- Lyapunov function $V(W)$ on the reduced flow — open build of the ACT companion paper [Smart, 2026a].
- $2I$-equivariance audit of the closure operator family — open build of ACT.
- Edge-space decomposition of $\mathbb{R}^{E_M}$ under the Hodge edge Laplacian — open build of ACT.
- Selection theorem identifying V_{600} over alternative regular 4-polytopes — see §9.1.
- Formal discrete-to-continuum convergence theorem under a specified large-graph limit — see §9.3.
- First-principles derivation of the φ^{-2} shift — see §9.2.
- Family-membership classification theorem for the polynomial-in-L Lyapunov family — see §8.

9.6 The honest verdict

The result is an *operator witness*: a geometry-fixed response operator on a fixed graph, with no shape parameters tuned to any dataset, is consistent with two domain-disjoint empirical landings in qualitatively distinct regimes. We do not claim the operator is the unique kernel for either landing. We do not claim selection is delivered. We do not claim 600-cell uniqueness. The strengthening builds for stronger claims are listed above and remain open.

10 Conclusion

The closure-response operator $C_\varphi = L_M + \varphi^{-2}I$ on the 600-cell graph V_{600}, with $\varphi = (1+\sqrt{5})/2$, is a geometry-fixed response primitive: positive definite under (H1)–(H3) on the substrate (M, L_M), and on the connected finite graph V_{600} where $\inf\sigma(L_M) = 0$ it has smallest eigenvalue φ^{-2} and operator-norm identity $\|C_\varphi^{-1}\| = \varphi^2 \approx 2.618$ (general substrates with $\inf\sigma(L_M) > 0$ give the strict inequality $\|C_\varphi^{-1}\| < \varphi^2$). The 600-cell instance has $|V| = 120$, $|E| = 720$, uniform degree 12, 9-shell partition $\{1, 12, 20, 12, 30, 12, 20, 12, 1\}$, and a Laplacian spectrum that numerically resolves to the closed-form $\mathbb{Z}[\varphi]$ values listed in Table 1. The discrete-to-continuum agreement between $\psi = C_\varphi^{-1}f$ and the continuum kernel $G(x) = (\varphi/2)\,e^{-|x|/\varphi}$ at per-vertex chord distances (non-source vertices) is Pearson $\rho = 0.976$ on the unweighted Laplacian, above the two φ-cocycle weighted variants tested (0.888 geometric, 0.884 arithmetic). All numbers are reproduced from canonical generators by `repro/verify_kernel.py`; no parameter is fitted.

Two domain-disjoint empirical landings. The same fixed C_φ on the same fixed V_{600}, with no shape-parameter retuning between regimes, appears as the shared response primitive underneath:

1. **Passive regime** [Smart, 2026c]: a single fitted dimensionless amplitude A per dataset (kernel shape held fixed) gives a sign-uniform $\Delta C_9^{\mathrm{eff}} < 0$ description of the $b \to s\mu^+\mu^-$ angular anomaly across five public datasets (LHCb 2015, LHCb 2021, CMS 2025, LHCb 2025, LHCb 2015 $B_s \to \phi$).
2. **Active regime** [Smart, 2026b]: a recurrent self-model layer above the same operator (one condition-dependent self-injection coupling $\eta \in \{0, 0.05, 0.20\}$, one substrate-pinned nonlinearity bounded_topk$(\cdot, k = 12)$ at the graph's average degree) is consistent with the aria-chess active-regime tally (frozen 2026-04-18): 17/18 preregistered substrate/neuroscience

correspondences under the standard validation protocol, 18/18 after the documented $N = 20$ P4 deep-dive (thresholds unchanged; with the documented P3 $N = 5$ and P13 LOO/state-reset refinements, and the open prereg-exact P10 20-permutation rerun, all inherited verbatim from [Smart, 2026b]), plus six literature-thresholded drug/sleep EEG signatures.

By design, the two witnesses share exactly the geometry-fixed operator: the same C_φ, substrate V_{600}, and shift φ^{-2}. Above that operator level, they share no fitted parameter, threshold, or dataset, and the empirical claims are tested on disjoint physical domains (flavour physics vs cortical neuroscience). Independence here is independence at the empirical layer above the shared operator, not statistical independence of the operator itself.

Operator-witness scope. This is an operator witness, not a derivation of physics on either landing. We do not derive the φ^{-2} shift; it is a design-level stability clamp. We do not claim 600-cell uniqueness; alternative regular 4-polytopes are an explicit ablation build. We do not claim kernel uniqueness on either empirical landing; the b-anomaly's free-width Gaussian alternative, AIC tie ($w_{\mathrm{VFD}} = 0.348$ vs $w_{\mathrm{FREE\_C9}} = 0.652$), and Mode-B linearised-to-non-linear refit drift caveats are inherited verbatim, and the aria-chess substrate-witness scope is inherited verbatim. We do not deliver a selection theorem on the ACT 4-tuple [Smart, 2026a]; that paper's open builds (Lyapunov $V(W)$, edge-space decomposition under $2I$-equivariance, full reduced-flow convergence) remain open and are not delivered here.

Programme position. §8 positions C_φ relative to the published adaptive-closure-transport companion as a member of the same polynomial-in-L family; family-membership is *programme-positioned*, not formally classified. The two empirical landings provide external consistency checks on the *response* primitive without reducing or addressing the selection gap.

The empirical material gathered here is the operator witness; the broader programme to convert the witness into a selection-theorem- grade claim is sketched in the companion preprints and remains the natural next step.

Reproducibility

The complete numerical verification (vertex construction, short-edge graph build, Laplacian spectrum, operator-norm bound, discrete-to-continuum correlation across three Laplacian variants) is reproducible from `repro/verify_kernel.py` in this paper's bundle. No randomness, no fitted parameters: all numbers in §3, §4, and §5 are deterministic outputs of the script. The two empirical witness preprints (b-anomaly [Smart, 2026c], aria-chess [Smart, 2026b]) carry their own reproducibility artifacts; this paper does not duplicate them.

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